

2024-25 Program-level Annual Assessment Report

This report should be a collaborative effort involving the program-level Assessment Coordinator, the evaluators and the Department Chair.

Program Name: MS in Computational Decision Science and Operations Research

Department Name: Computer Science

College Name: College of Science

Faculty Assessment Coordinator:

1. PROGRAM LEARNING GOALS EVALUATED IN THIS ASSESSMENT CYCLE:

List the program learning goals that were evaluated in this assessment cycle. Do not simply reference the assessment plan for this program.

1. Knowledge of Design and Analysis of Algorithms
2. Knowledge of Optimization Methods
3. Knowledge of Statistics

2. ASSESSMENT METHODOLOGY: Use the table below to describe your assessment methodology. Do not simply reference the assessment plan for this program. *Copy the table for each learning goal assessed in the last academic year.*

First Learning Goal		
Learning Goal Assessed	G1: Knowledge of Design and Analysis of Algorithms	
Sample size		
Semester(s) in which artifacts were collected	SPRING-FALL 2024 21	
Name of rubric used to evaluate student artifacts	Algorithmic Technique Rubric (See attached)	
Artifact source	Course(s) and Instructor(s): CS430/ CS535	Assignment(s):
	Other (specify):	
Month/Year of Assessment/Evaluation	FALL 202 5 2	
Names & Titles of the evaluators	Sanjiv Kapoor, Professor Computer Science	

Second Learning Goal		
Learning Goal Assessed	G2: Knowledge of Optimization Methods	
Sample size		
Semester(s) in which artifacts were collected	FALL 2024 /SPRING 202 5 2	
Name of rubric used to evaluate student artifacts	Optimization Methods-Rubric	

Artifact source	Course(s) and Instructor(s): MATH535 /CS538	Assignment(s):
	Other (specify):	
Month/Year of Assessment/Evaluation	FALL 2025	
Names & Titles of the evaluators	Prof. Hemanshu Kaul / Sanjiv Kapoor	

Third Learning Goal		
Learning Goal Assessed	G3: Knowledge of Statistics	
Sample size		
Semester(s) in which artifacts were collected	FALL-SPRING 2025	
Name of rubric used to evaluate student artifacts	Statistics Assessment Rubric	
Artifact source	Course(s) and Instructor(s): MATH564/ MATH565	Assignment(s):
	Other (specify):	
Month/Year of Assessment/Evaluation	FALL 2025	
Names & Titles of the evaluators	Prof. Hemanshu Kaul	

3. ASSESSMENT RESULTS: Insert a table or graph summarizing the results. Results should be presented by performance indicator for each learning goal. If the data were collected in Blackboard Outcomes, the IIT Assessment Office will provide the information to insert into this section of the report (see samples below).

(1) Learning Goal G1:

Ability to describe a problem mathematically and design, analyze algorithms and characterize the complexity of problems.

Rubric for Learning Goal G1:

Topic	Percentage Satisfying Goal
1. Ability to Solve recurrence relations	80% (5/6 achieved 70%) <u>100%</u>
2. Knowledge of algorithmic techniques listed below (a) Divide and Conquer (b) Dynamic Programming (c) Greedy Method	80% (5/6 achieved 70%) <u>85%</u>
3. Ability to evaluate the complexity of algorithms	100% (6/6 achieved 70%)
4. Ability to create Graph models and apply algorithms to solve problems on Graphs	80% (5/6 achieved 70%) <u>85%</u>
5. Knowledge of lower bounds and Complexity Classes.	100% (6/6 achieved 70%) <u>Not Tested in CS430</u>

(2) Learning Goal G2:

Ability to create and solve mathematical models of problems, using combinatorial optimization and mathematical programming techniques; and evaluate the complexity of the solution.

Rubric for Learning Goal G2:

Topic	Percentage Satisfying Goal
1. Ability to formulate optimization problems as mathematical models, examples including linear programming and combinatorial models, e.g. ability to solve other problems using Graph Matchings and Network Flows (when possible) and Linear programming. hw1	85% (6/7) achieved 85% <u>100%</u>
2. Knowledge of Linear programming and Integer Linear Programming. Develop insight into the geometric basis and linear algebra solution of linear programs. hw3	85% (6/7) achieved 85% <u>100%</u>
3. Knowledge of algorithms that provide (i) Solutions for linear programming (ii) Solutions for problems in combinatorial models, e.g. Graph Matchings and Network Flows-hw4,5	100% (7/7) achieved 85% Data for transfer students not available <u>85%</u>
4. Knowledge of duality and its role in solving combinatorial models and linear programming. hw2	85% (6/7) achieved 85% <u>85%</u>

(3) Learning Goal G3:

Ability to use regression and forecasting models to statistically model processes and data from various fields of science and engineering.

Rubric for Learning Goal G3

Topic	Percentage Satisfying Goal
1. Ability to articulate the assumptions of regression and forecasting models, and understanding the consequences of having these assumptions violated.	100% (6/6 achieved 70%) 67%
2. Ability to evaluate the adequacy of fitted models using residual plots and statistical tests, determine whether the data violate any of these assumptions and propose remedies for assumptions that are violated.	100% (6/6 achieved 70%)
3. Ability to use statistical software to fit regression and forecasting models to real data, and use the associated output from this software to evaluate the model fit to the data.	100% (6/6 achieved 70%)
4. Ability to communicate the statistical analyses of substantial data sets through explanatory text, tables and graphs.	100% (6/6 achieved 70%)
5. Ability to detect autocorrelation of time series data, using graphical and statistical methods, and propose corrective measures.	100% (6/6 achieved 70%)

4. DISCUSSION OF RESULTS: Use this section to describe the key findings and program performance issues revealed in the interpretation of the data. *The evaluators should provide input into this section of the report.*

Goal G1: Students were able to meet the goals by the end of their studies in CS430 /CS535 (Algorithm analysis and Design). It is recommended to discuss improvements on the goal of analyzing underlying mathematical analysis.

Goal G2: Students were able to meet the sub-goals.

Goal G3: Students were able to meet almost all the sub-goals.

5. IMPROVEMENT PLANS: Use this section to provide specific information about what elements of the curriculum may need to be modified in order to improve the program's performance. *This section should be completed and signed by the Department Chair.*

Specific modification to courses or assignments or curriculum.	Name and title of person responsible for implementing the changes.	Date by which changes will be in place.	Intended result
Discussion with instructors of CS 430/CS535 (if CDSOR students are taking the courses)	Sanjiv Kapoor	December 2025	Check in Performance:
Discussion with instructors of CS538/MATH535, if CDSOR students are taking the course.	Hemanshu Kaul	December Spring 2025	Improving Performance.
Discussion with instructors of MATH564/MATH565, if CDSOR students are taking the course	Hemanshu Kaul	December Spring 2025	Discuss performances.

Department Chair should sign below:

Department Chair Name

Signature

Date

6. ASSESSMENT PROCESS RECOMMENDATIONS: Use this section to provide feedback on the assessment process itself.

The assessment process of CDSOR is currently course based. As more students take projects measurement of project performance will be considered. We had well-trained students in the program from the Europe joint program.

7. ASSESSMENT PLAN FOR 20242-253: Use the table below to describe your assessment plan for the current academic year. Do not simply refer to an assessment plan submitted previously. *Copy table for each learning goal to be assessed.*

First Learning Goal

Learning Goal Assessed	Knowledge of Design and Analysis of Algorithms	
Expected Sample size	5	
Semester(s) in which artifacts were collected	FALL 20252/ SPING 20263	
Name of rubric used to evaluate student artifacts	Algorithmic Technique Rubric G1 (See above)	
Artifact source	Course(s) and Instructor(s):	Assignment(s):
	CS430/CS535	
	Other (specify):	
Month/Year of Assessment/Evaluation	MAY 20263	
Names & Titles of the evaluators	S. Kapoor	

Second Learning Goal		
Learning Goal Assessed	Knowledge of Optimization Methods	
Sample size	unknown	
Semester(s) in which artifacts were collected	SPRING 20263	
Name of rubric used to evaluate student artifacts	Optimization Knowledge Rubric (G2)	
Artifact source	Course(s) and Instructor(s): CS538/MATH535	Assignment(s):
	Other (specify):	
Month/Year of Assessment/Evaluation	MAY 20263	
Names & Titles of the evaluators	H. Kaul	

Third Learning Goal		
Learning Goal Assessed	Knowledge of Statistics	
Sample size	3	
Semester(s) in which artifacts were collected	FALL 2022 / SPRING 2023	
Name of rubric used to evaluate student artifacts		
Artifact source	Course(s) and Instructor(s): MATH 564/ MATH 565	Assignment(s):
	Other (specify):	
Month/Year of Assessment/Evaluation	MAY 2023	
Names & Titles of the evaluators	H. Kaul	

8. DEAN'S REVIEW: The Dean should use this space to comment on each of the proposed curriculum changes.

List of specific modifications to courses or the curriculum.	Dean's Response

9. REPORT SUBMISSION: Please submit this report to Carol Emmons (emmons@iit.edu) by *October 15, 2022*. For questions about the completion of this report, email: emmons@iit.edu or call 312-567-3827..

Name of person submitting report	Date submitted